



Muhammad Irfan bin Azman

Piping Engineer / Jr Pipe Stress Engineer

Summary

Piping Engineer in the oil & gas sector with 4 years of experience (including 2 years in piping engineering), bringing practical expertise in piping design, 3D modelling, Pipe Stress and layout creation. Skilled in interpreting P&IDs and isometric drawings, with knowledge of ASME B31.3 and related industry codes. Proficient in Smart 3D, Navisworks, and AutoCAD for developing and validating piping systems that meet project specifications. Experienced in leveraging Power BI daily for data analysis, dashboard creation, and performance monitoring to support project decision-making. Holds a BEng in Mechanical Engineering from UTHM, with a strong focus on safety, quality, and continuous professional growth in dynamic engineering environments.

Education

2018-2020 Bachelor's Degree in Mechanical Engineering, Universiti Tun Hussein Onn, Malaysia
2014-2017 Diploma in Mechanical Engineering, Universiti Tun Hussein Onn, Malaysia

Experience

2024-Present Piping Design / Jr Pipe Stress Engineer, Ranhill Worley, Kuala Lumpur

Feb 2024-Present SEATRIUM / PETROBRAS, DED P82 FPSO, Kuala Lumpur

- Jr Pipe Stress Engineer
 - Perform CEASER II stress modelling in accordance with ASME B31.3
 - Extracted loads (Hanger, Nozzle, Restrain) from CESSER II for pipe support design
 - Coordinated with stress engineer to implement piping flexibility solutions and modify routing/supports during review cycle
 - Identified overstress and support clashes, proposing revisions for compliance with project standard
 - Cross-checked pipe stress reports against 3D models and isometric drawings, marking up supports to align with the latest stress report
- ISOMETRIC Coordinator
 - Coordinated and managed the issuance of isometric drawings for piping systems, ensuring compliance with project specifications and mechanical design standards.
 - Reviewed and verified isometric drawings for accuracy against engineering requirements, maintaining data integrity throughout the project lifecycle
 - Monitored and enforced strict weekly deadlines for drawing issuance, supporting project schedule adherence.
 - Maintained detailed documentation of drawing revisions and re-issuance timelines for internal tracking and audits.

- Supported Lead and Senior Engineers with data consistency checks across design deliverables.
- Developed and presented interactive dashboards in Power BI to provide clients with real-time project insights, drawing progress, and data summaries..
- Other task
 - Reviewed P&ID and 3D models to identify line changes against issued and pending isometrics.
 - Assisted in preparing tubing MTO (Material Take-Off) to support procurement and construction activities.

2023-2024 Graduate Proposal Engineer, Ranhill Worley, Kuala Lumpur

Mar 2023- Jan 2024

- Graduate Proposal Engineer
 - Prepared complete technical and commercial proposals for upstream oil & gas packages valued up to RM 15 million.
 - Interpreted client RFQs and applied relevant codes to produce compliant equipment lists and technical data sheets.
 - Developed detailed cost estimates and cash flow curves to support bid evaluations
 - Coordinated multi-discipline inputs (engineering, procurement, tax, legal, treasury), ensuring 95% of submissions delivered on or ahead of schedule.
 - Led both technical (execution plans, project team structures) and non-technical activities to ensure comprehensive proposal coverage.
 - Consolidated inputs from diverse stakeholders into clear, competitive submissions.
 - Contributed to securing two project wins (CPOC FEED & Shell MDS reimbursable contract basis).

2022-2023 Project Engineer, Trengco Energy (M) Sdn Bhd, Kerteh

Mar 2022-Feb 2023

- Planned, executed, and controlled projects in compliance with quality and safety standards
- Analyzed engineering drawings, specifications, and purchase orders to ensure project requirements were met
- Prepared execution documentation including work instructions and operating procedures.
- Supported the Project Manager in negotiating change orders, coordinating subcontractors, and ensuring smooth project execution.
- Conducted site visits during pre-bid stage to evaluate project scope and risks.
- Organized cross-functional meetings to resolve issues, align stakeholders, and maintain project control..
 - Key Project
 - PETRONAS Penapisan Terengganu (PPTSB): Cleaning & desludging, waste treatment, and civil works.
 - PETRONAS Chemicals Olefins, Glycols & Derivatives (PCOGD): Amine pump replacement.
 - Arkema Thiochemicals Kerteh: Fabrication and installation of actuated valve supports.

2021-2022 Mechanical Rotating GEES, PETRONAS Gas Berhad, Kerteh

Apr 2021-Mar 2022

- Assist engineer to monitor healthiness and performance of rotating equipment at Utilities Kerteh and Utilities Gebeng using special monitoring and prediction system for rotating equipment (MMPD)

- Prepare pre-emptive work control sheet for boiler feed water pump, cooling tower (gearbox & fan blade)
- Involve in Major Inspection (MI) for gas turbine (GT-B) at utilities Kerteh – GE MS 6001B GT
- Involve in refurbishment work for MRO Pump at Utilities Kerteh

2020-2021 Asst Project Engineer, TMA Construction, Paka

Apr 2020-Mar 2021

- Plan and coordinate in site activity for EPCC NAP (New Air Compressor Project) at PGB (GPP 5&6), earthwork and site preparation.
- Prepare daily progress report for main contractor (CMI)
- Prepare and review document for site inspection.
- Attend site visit prior project bidding.

Memberships and affiliations

2020	Board of Engineer (BEM), Graduate Engineer Mechanical – G1199436A
2020	IEM, Member of Institutes of Engineer Malaysia

Technical courses

2020	NIOSH, Oil and Gas Safety Passport (OGSP)
2020	NIOSH, Authorised Entrant and Standby Person for Confined Space (AESP)
2020	CIDB, Green card

Specialist and technical skills

- Intergraph Smart 3D
- CAESER II
- AutoCAD
- Power BI
- SolidWorks
- Navisworks
- Microsoft Office
- Pipedata-Pro

Languages

- Malay - Natives
- English – Professional working proficiency

Contact

- Phone: 017-941 3702
- Email: irfanazman96@gmail.com
- Address: Jalan Sungai Besi, Kuala Lumpur

Reference

SAIFUL EDZUAN

DEPUTY HOD PIPING ENGINEERING
RANHILL WORLEY
019-660 0776
SAIFUL.EDZUAN@WORLEY.COM

RAJENDRAN RAMALETCHUMAN

SENIOR PIPING ENGINEER
MCDERMOTT MALAYSIA
012-216 4626
RAJ_EN85@YAHOO.COM.